Subnet Planning Assignment

1. Division of 192.168.10.0/24 into Four Equal Subnets

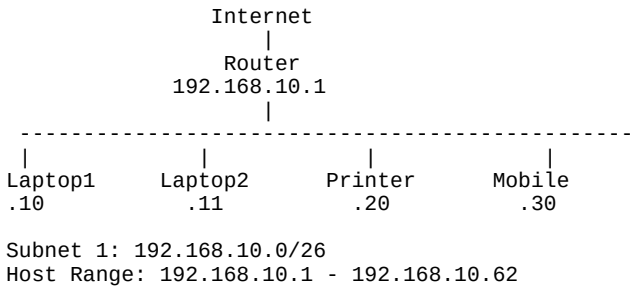
```
Subnet Mask: /26 (255.255.255.192)

Subnet 1: Network 192.168.10.0 | Broadcast 192.168.10.63
Subnet 2: Network 192.168.10.64 | Broadcast 192.168.10.127
Subnet 3: Network 192.168.10.128 | Broadcast 192.168.10.191
Subnet 4: Network 192.168.10.192 | Broadcast 192.168.10.255
```

2. Device IP Assignments (Subnet 1)

```
Router      : 192.168.10.1
Laptop 1    : 192.168.10.10
Laptop 2    : 192.168.10.11
Printer     : 192.168.10.20
Mobile      : 192.168.10.30
Default Gateway: 192.168.10.1
```

3. Network Diagram




```
Subnet 1: 192.168.10.0/26
Host Range: 192.168.10.1 - 192.168.10.62
```

4. Explanation of Calculations

Starting from a /24 network, two host bits are borrowed to create four equal subnets. This results in a /26 subnet mask (255.255.255.192). Each subnet contains 64 addresses, of which 62 are usable by hosts. Device addresses were selected from the usable host range while avoiding the network and broadcast addresses.